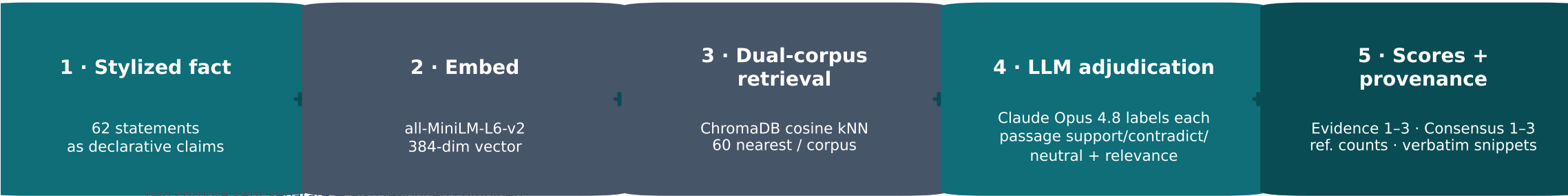


AI-Assisted Assessment of Stylized Facts — Method & Validation

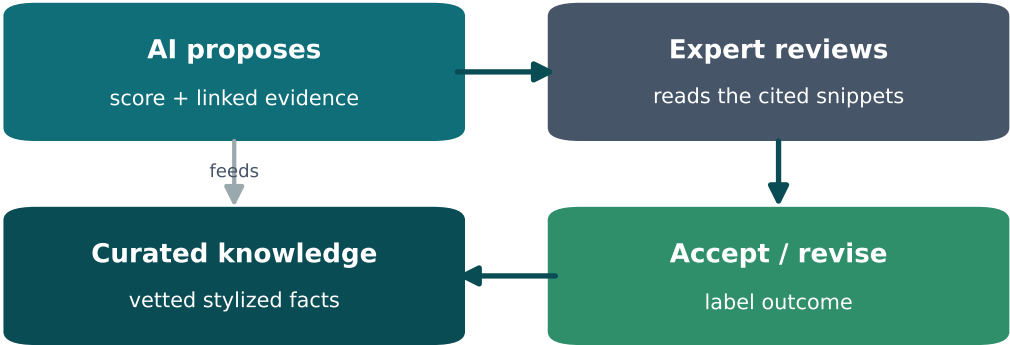
Retrieval-augmented, LLM-adjudicated scoring of empirical patterns, with a human-validatable, fully traceable design

METHOD



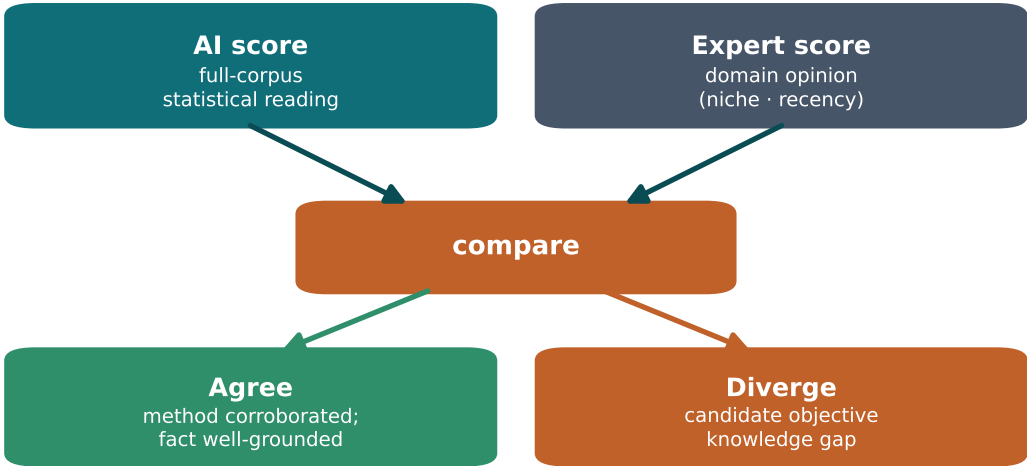
VALIDATION

1 · Human-in-the-loop curation



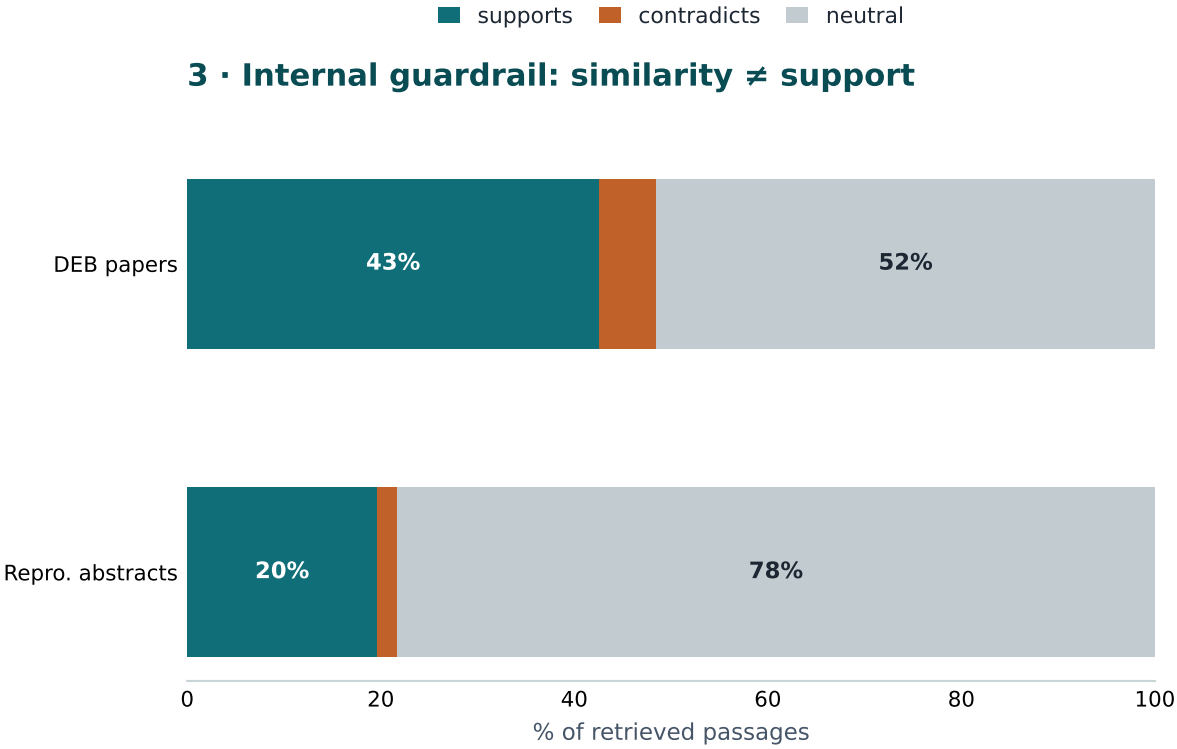
every step recorded → transparent audit trail

2 · AI ↔ expert comparison



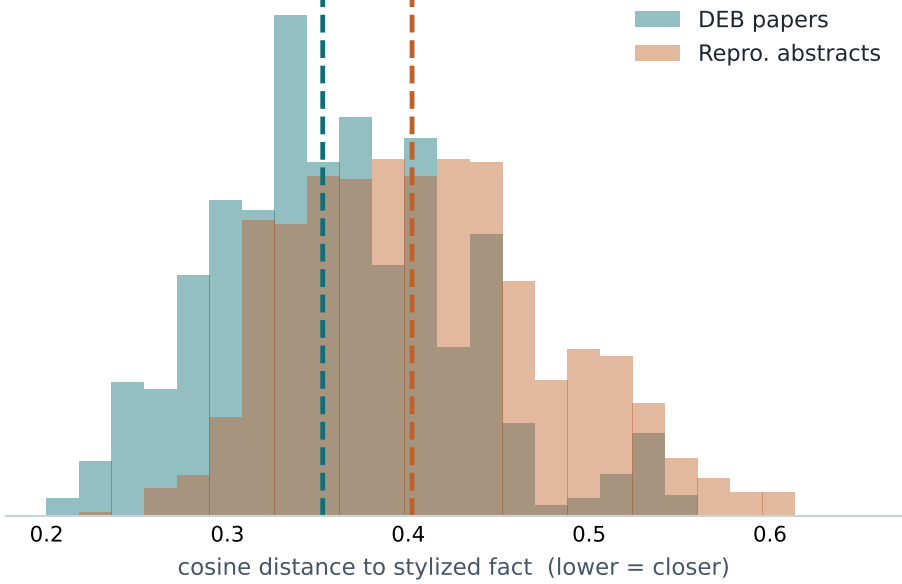
empirical agreement reported in main text

3 · Internal guardrail: similarity ≠ support



Even among the nearest passages (cos-dist < 0.45), 60% are judged NEUTRAL — the adjudication step, not similarity, decides support.

4 · Retrieval quality



Dashed = median. The focused DEB corpus sits closer to the facts than the broad abstract corpus, as expected.

Traceable	Adjudicated	Reproducible	Gap-finding
every score links to its source passages	60% of nearest passages filtered out as neutral	deterministic retrieval; fixed parameters	AI-expert divergence flags knowledge gaps to probe

Validation philosophy. The system is designed to be checked, not trusted blindly: similarity locates candidate text, but an explicit support/contradict adjudication — auditable down to the verbatim passage — decides evidence. Because the AI reads entire corpora statistically whereas an expert judges from domain experience, agreement corroborates a fact while disagreement is treated not as AI error but as a signal worth investigating: a candidate, objective knowledge gap that a reasoning LLM agent (or a human) can target. The empirical AI-versus-expert comparison is reported in the main text.